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Royal Government of Bhutan
Ministry of Agriculture and Forests
Department of Forests and Park Services



MANAGEMENT AND MARKETING PLAN FOR NON- WOOD FOREST PRODUCTS

(Goney-Wambur Shingmin Tshokpa)



Tsenkhar Gewog
Lhuntse Dzongkhag

(Plan Period: May 2022-May 2027)

Approval Sheet

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PART A: MANAGEMENT AND MARKETING PLAN

1.1 Introduction

Bhutan is blessed with many NWFPs wherein the local community use it for self-consumption as well as for generating income since the time immemorial. The NWFPs as such plays vital role in improving livelihood of the people residing near and far from the forest. However, these resources are often collected from common pool and potential for conflicting issues. Therefore, Department of Forest and Park Services have been helping the general public in managing resources and making people to use the resources sustainably without compromising to its conservation.

Goney-Wambur is a chiwog under Tsenkhar Gewog which is located approximately at an elevation of 1300-2000 masl. The chiwog is connected with feeder road, telephone, electricity and clean drinking water supply. It is 23 km drive from the Gewog Centre, Phawn. The village experience warm summer and cold winter. The chiwog has 73 households of which 63 households are member in the NWFP management group. They are mostly agrarian society depending on agricultural farming for their daily livelihood. Besides agricultural farming, people in the community rear also livestock and also engaged in forestry activities. They practice subsistence farming and main staple food crop grown in the area are rice, maize and wheat etc. They also produce a wide variety of vegetables such as chili, potato, cabbage, radish and beans etc.

The forest type surrounding the community is conifer and cool-broadleaved forest dominated by the cool-broadleaved forests. The vast stretches of thick and dense forest provides home to wide range of wild animals such as Wild Boar, Mask Deer, Barking Deer, Bear, Common Leopard, Fox, Dhole, Monkey and variety of birds species.

The non-wood products are distributed in small patches of the area in limited quantity. There are total of 13 Non Wood Forest Products (Table 1) found in the designated area and out of which, 4 has economically potential viz. ***Paris pollyphylla*** (Jagpa Thogsumpa), ***Daphne bholua***, ***Rubia cordifolia***, ***Borinda grossa***. Although huge quantity of NWFPs may not be available in first few years, but has potential for bountiful harvest through application of appropriate management regimes and regulation of collection by community.

Approximately more than 5934 acres of adjoining forest area provides conducive area for the community people to collect Non-Wood Forest Products for the Member of **Goney-Wambur Shingmen Tshogpa**.

1.2 Objectives of Collection and Marketing of NWFP:

1. Sustainable management and harvesting of Non Wood Forest Products.
2. To generate additional income from sale of Non-wood Forest Products.
3. To protect collection of NWFP Products by outsiders.

1.3 Description of NWFPs:

The forest of Goney-Wambur has varieties of NWFPs such as *Paris pollyphylla* (Jagpa Thogsampa), *Borinda grossa* (Baa), *Rubia cordifolia* (Tsoe), *Daphne bholua* (Deynap), *Onamas hookari* (Muhtsi), *Pogostemon amaranthoides* (Namda), *Diplzium esculatum* (Fern), *Cymbidium spp* (Olachoto), *Acorus calamus* (Shuda), *Elantosteme lineolatum* (Damroo), *Amblica officialnalis* (Amla), *Nephrolepsis cordifolia* (Khowchi khiwchi), *Viscum nepalensis* (Ngashing Jorma), *Otochilus lancilabius* (Olatse). The most of the NWFPs are found in upper ridge of Wambur and Lagyel village which is dominated with broadleaved forest while Amla and Khowchi is found in lower elevation especially in Chir-pine dominated forest. The Baa and Deynap is found abundantly in Lagyel site and local people extract Baa for both domestic and commercial purposes. Although Daphne economic value in other areas people have not collected so far. This has a potential to become next NWFP to earn additional income. Presently, the group mainly focuses on those economically viable NWFPs such as *Borinda grossa*, *Rubia cordifolia*, and *Paris polyphylla* as the main products for commercial marketing where *Daphne bholua* will be collected only if there is demand from the buyers/traders. NWFPs such as ferns and mushroom will be collected mainly for domestic purposes.

Table 01: Brief description of NWFP for commercial harvesting

NWFP species	Harvesting area(s) name	Part harvested	Use purpose(traditional & others)	Harvesting season(months)	Estimated quantity that can be collected in a year
<i>Borinda grossa</i>		culms	Weaving bamboo mates/handicraft	November to February	1000culms
<i>Rubia cordifolia</i>		Upper part of plant		October to March	3000kg

Paris polyphylla		Rhizome		October to November	200kg
Daphne boula		Bark			7000kg

Table 02: Harvesting prescription for sustainable harvesting of NWFP

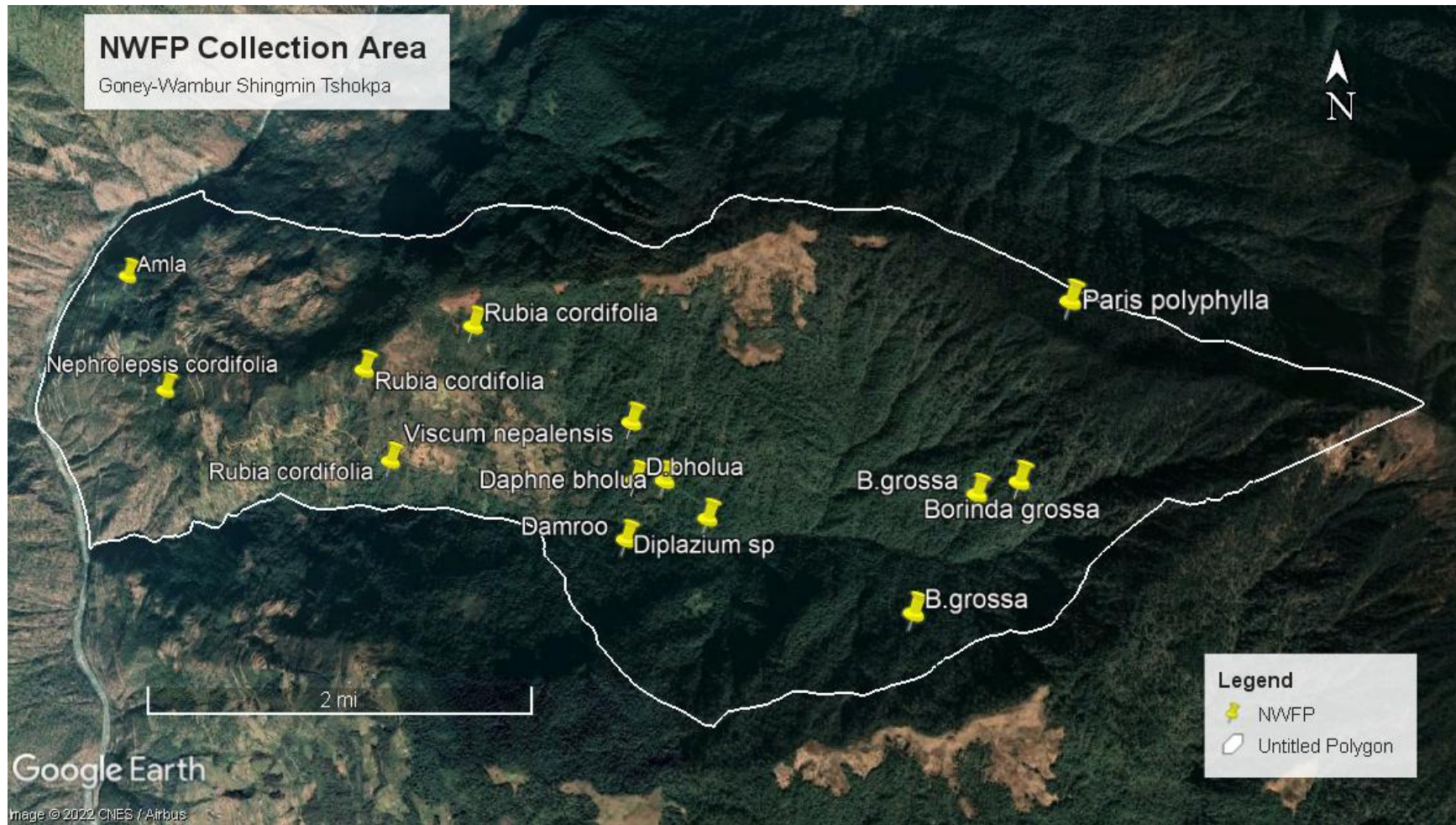
NWFP species	Harvesting prescription for sustainable harvesting
Borinda grossa	
Rubia cordifolia	
Paris polyphylla	
Daphne boula	

Table 1: Harvesting Guidelines and Description

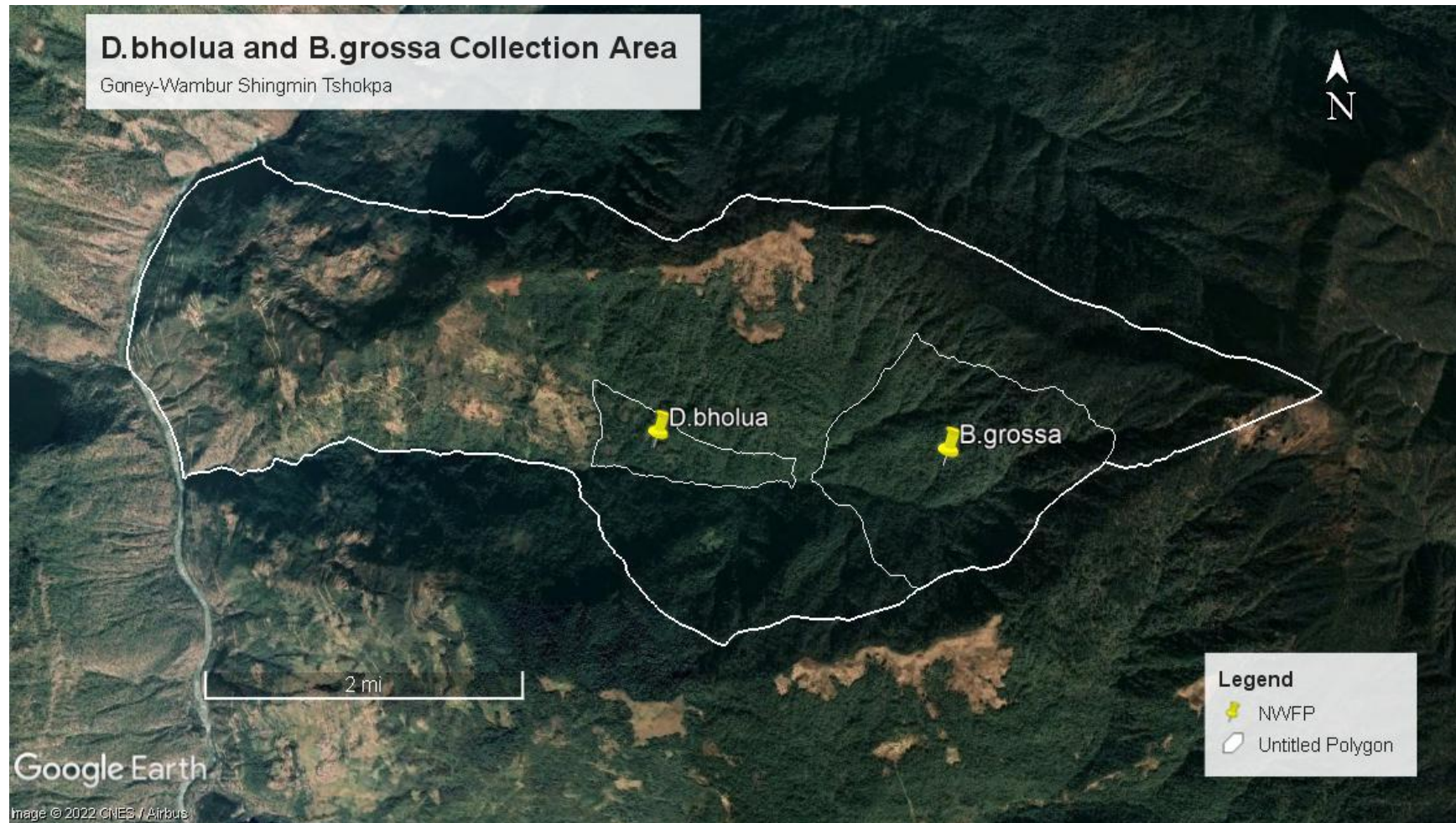
Sl.No	Name		Parts Used	Purpose	Time of Harvesting (month)	Harvesting Guideline
	Scientific	Local				
1	<i>Paris pollyphylla</i>	Jagpa Thogsupa	Root/Rhizome	Medicine	October to November	Dig up only mature rhizomes. Leave the mother plant intact with small part of root in the soil. Harvest rhizome on during harvesting season. Do not uproot entire plant.
2	<i>Borinda grossa</i>	Baa	Stem/culms	Mates and other products	November to February	Cut culm at the base or at first long internodes above the nodes. Leave bamboo shoot or culm of less than one year old intact. Do cut culm when new shoots are sprouting. Harvest culm in condense culm in u shape
3	<i>Rubia cordifolia</i>	Tsoy	Root.stem n leaves	Medicine	October to March	Use sharp scissors for the harvesting of the upper parts of the plants. The plant has to be cut from 5cm above the ground. Do not uproot entire plants. Leave at least 10% of the plants without collecting to ensure natural regeneration.
4	<i>Dephna bholau</i>	Desho	Bark	make traditional paper	February to September	Cut the plant 10cm above ground level. Cut stem that are more 3cm in diameter. Do not damage the remaining young stands/seedlings or saplings.

5	<i>Onamas hookari</i>	Muhtsi	Roots	religiouse purpose	September to December	Dig up only mature rhizomes. Leave the mother plant intact with small part of root in the soil. Harvest rhizome on during harvesting season. Do not uproot entire plant.
6	<i>Pogostemon amaranthoides</i>	Namda	leaves	vegetable	May to September	Cut tender shoot only. Retain 1% of mother plants in clump
7	<i>Diplazium sp</i>	Fern	shoot	vegetable	June-September	Cut tender shoot only. Retain 1% of mother plants in clump
8	<i>Cymbidium sp</i>	Olachoto	Tender parts	Vegetables	7-8 and 8 Bhutanese month	Cut tender shoot only.Retain mother plants in clump
9	<i>Acorus calamus</i>	Shuda	Roots	Medicine	whole year	The plant is uprooted either by hand or using hose,cut the root parts and plant back the aerial stem
10	<i>Elantosteme lineolatum</i>	Damroo	leaves	Medicine and food	9 th to 12 Bhutanese month	Cut tender shoot only
11	<i>Emblica officialnalis</i>	Amla	leaves	medicine/food	January to March	pluking by hand or shaking with small stick Picking fruits from the forest floor. Dot cut down tree inorder for fruits to collect.
12	<i>Nephrolepsis cordifolia</i>	Khowchi khiwchi	root	medicine	whole year	The plants are uprooted either by hand or hoe. Retain few shoot as mother trees
13	<i>Viscum nepalensis</i>	Ngashing Jorma	shoot		March to May	The plants has to cut 20cm above ground (retain few plants as seed trees)

1.3 Map of the NWFP harvesting areas



1.4 Map of *B.grossa* and *D.bholua* extraction



1.5 Sustainable NWFP harvesting prescriptions

1.5.1 Domestic purpose

Any individuals harvesting NWFPs for domestic consumption should inform the Chairperson before proceeding for collection.

Harvesting of NWFPs for domestic purposes can be done any time with the realization of Royalty. Meanwhile, members shall comply with the restrictions (Reedam/Laadam) mandated by the village.

Maximum harvesting limit of *Borinda grossa* for domestic purposes is 200 numbers per household/season.

NWFPs like mushroom, damroo and fern for domestic consumption can be harvested anytime as per the requirement of the individual members.

1.5.2 Commercial purpose

Commercial harvesting should be done together involving all the members after obtaining collection permit from nearby forest office.

Any members willing to harvest the *Paris polyphylla*, *Rubia cordifolia* and *Borinda grossa* for commercial/domestic purpose, should inform the Chairperson before collection so that the chairperson could process for obtaining the collection permit.

The commercial extraction of NWFPs shall be done only by the members; the outsider is strictly restricted for commercial extraction.

1.6 Marketing and sales plan

- The NWFPs will be sold to the buyers/traders by the group at least involving one fourth of the member.
- No individual shall be permitted to sell without the consent of the group members.
- Standard/Harvestable size (2-4 cm) and length (5 m) of *Borinda grossa* for commercial purposes have to maintain. One back load should consist of 30-40 nos. for men and 15-20 nos. for women and children. It shall be sold at a rate of Nu.6/- which includes the royalty for outsiders and Nu.2 for group members.

- One kg of dried *Daphne bholua* shall be sold at Nu.50 including the royalty irrespective of quality.
- Cost of *Paris polyphylla* has been fixed at the rate of Nu.2500/ kg including the royalty.
- The dried *Rubia* shall be sold at the rate of Nu.50 per kg and shall not sell the fresh *Rubia*.
- Permit fee and royalty of NWFPs should be paid as per the royalty of the DoFPS.
- The other NWFPs listed in the annexure shall be sold as per the commercial rate fixed by the DoFPS.
- All the NWFPs shall be sold to the license holder only.

The marketing and sales plan for different types of NWFPs that are to be collected by the group, its potential markets where the group shall sell products with approximate quantity, price per unit and the total value of the expected sales are described in the table below:

Table 2: Marketing and financial projections

<i>NWFP</i>	<i>Potential market</i>	<i>Estimated quantity for sale per season (unit)</i>	<i>Expected price per unit</i>	<i>Total (Nu)</i>
Borinda grossa	Local Market	10000 culms	6	12000
Rubia cordifolia	Within Bhutan/India	3000 kg	50	50000
Paris polyphylla	Within Bhutan	200 kg	2500	250000
Daphne sp	Local market	7000 kg	50	35000
<i>Pogostemon amaranthoides</i> -Namda	Local market	400 bundles	20	2000

<i>Diplazium esculentum</i> -Fern	Local market	800 bundles	20	3000
<i>Elantosteme lineolatum</i> - Damroo	Local Market	400 bundles	15	2250
<i>Emblica officialnalis</i> - Amla	Local market	700 kg	50	5000
<i>Nephrolepis cordifolia</i> - Khowchi	Local Market	100 kg	80	4000
<i>Onosma hookeri</i> - Mutsi	Within Bhutan	100 kg	100	1000
<i>Cymbidium spp</i> - Olachotoh	Local Market	300 bundles	150	15000
<i>Viscum nepalensis</i> - <i>Ngashing Jorma</i>	Local Market	250 kg	80	4000
<i>Acorus calamus</i> - <i>Shuda</i>	Within Bhutan	150 kg	50	2500
Total Amount				1215500

Note: The marketing and sales plans for NWFPs list in the above table (qty) is estimated for 5 years.

1.7 Marketing arrangements

The Chairperson shall collect demand of NWFPs before onset of any product collection season with support of concerned Forestry Officials. Then, the marketing of NWFPs will be organized and arranged as per the procedures given below:

- Marketing of products will be always group based. However, the chairperson will arrange selling point either at Lhuntse, Mongar or in nearby villages and towns. The members will collect vegetable items from different blocks on rotational basis. It will be taken to vegetable market along with other agriculture products. OR
- Members are not allowed to accept any advance payment from buyers at any circumstances as such practices deteriorate quality of products and also encourage over exploitation of resources.

1.8 Resource Assessment

The Resource assessment of two potential NWFP viz. *Borinda grossa* and *Daphne bholua* in the area has been carried out to estimate the amount that can be harvested on sustainable basis from the area.

1.8.1 *Borinda grossa*

The culms of *Borinda grossa* is long, straight and it is useful for weaving durable fencing mats and roofing mats, baskets, containers, hats.

Field data collection

Borinda grossa resource assessment was carried at Lagyel area following the stratified sampling method in which area is divided into different blocks according to the location and density. The plant is measured randomly in the chosen sample plots. The quantitative assessment was carried out by laying the sample plots of 10 x 25 m along the slope direction. From each sample plot, the bamboo stock is measured including the condition of the clump, number 1+ year, 2+ year and 3+ year culms and the number of culms that the local people consider harvestable. These harvestable culms are mostly 1+ year, 2+ years are in good health and are straight, have length of five m and a diameter of 2 to 4 cm. Resource assessment was done by collecting the data about the locations and quantities of species present in the area. Based on the results of this survey, the area is divided

in to 3 blocks according to the availability of the resources on the Google map. Out of 3 blocks, the farmers has decided and agreed to collect the *Borinda grossa* from the block alternatively. The members also decided to collect *Borinda grossa* based on AHL and as per the guidelines.

Data Analysis

Data collected from the sampling plots were entered in the excel sheet and then sorted cleaned for further analysis work.

Calculation of Annual Harvesting Limits (AHL)

AHL: 0% of the shoots +33% of the total stock of 1+ year and 2 + years old culms+70% of the 3+ years old culms (Table.....).

Based on the results of the inventories, the total and harvestable number of culms were calculated for each sub block. The AHL for 1+ year and 2+ years old culms is set at 33% of standing bamboo. This will ensure the regeneration of the species because the shoots of less than 1 year old culms will be retained and provide the 1+year culms for following year. There will be also enough 1 + year culms to provide 2 + years old culms for next season. Meanwhile, 70% Of 3 + years culms can be harvested and some will be left in the area to provide physical support to the younger culms. Based on the above AHL calculations, people decided to follow the following conditions;

1. Maximum of 33% of 1+ year and 2+ year old culms per clumps will be harvested and 67% will be left intact in the area for regeneration.
2. 70% of 3+ years old culms will be harvested and rest will be left in the area to provide physical support to the younger culms.

Table 3: Data analysis of *B. grossa*

Resource assessment compilation form for Borinda grossa						
Block name:	Lagay Bamboo	Dates of bamboo	26-28/01/2022			
Sub-block (Number)	Name of the sub-block	Area (ha)	% of productive	Productive area (ha)	Number of sample	Remarks
1	Rewapang Juug	94.5	34.4	32.5	5	
2	Sadug Brangsa	108	34.6	37.4	6	
3	Rashu Goenpa	83.3	27.1	22.6	6	
Age class	Number of culms	Sub-block 1	Sub-block 2	Sub-block 3	Total in Block	
1 + year old bamboo culms	Total counted in sample plots	362	729	298	1389	
	Total harvestible in sample plots	218	460	208	886	
	Total in sub-block*	94120	181764	44899	320783	
	Total harvestible in sub-block*	56680	114693	31339	202712	
2 + years old bamboo culms	Total counted in sample plots	405	743	285	1433	
	Total harvestible in sample plots	294	492	179	965	
	Total in sub-block*	105300	185255	42940	333495	
	Total harvestible in sub-block*	76440	122672	26969	226081	
3 + years old bamboo culms	Total counted in sample plots	591	791	335	1717	
	Total harvestible in sample plots	111	266	119	496	
	Total in sub-block*	153660	197223	50473	401356	
	Total harvestible in sub-block*	28860	66323	17929	113112	
Total Stock in Sub-Block		353080	564241	138312	1055633	
Total Harvestible in Sub-Block		161980	303688	76237	541905	
AHL-33% (1 & 2 + years)		136360	245237	57757	439353	
AHL-70% (3+ years)		107562	138056	35331	280949	

1.8.2 *Daphne bholua*

Daphne bholua (Deynap) is a shrub of 1-3.5 m in height and occurs in evergreen oak, spruce, hemlock, fir and blue pine forests at altitude of 1950-3400 m. It is used for the production of traditional paper. *D. bholua* is the only *Daphne* species that is also used as an ornamental plant because of its strong fragrant flowers.

Field Data collection

The resource assessment for *D.bholua* is carried out at Goney and Lagyel area following a systematic sampling design with the sampling plots of 10x10m. In each sampling plot only the diameter of the stem was recorded. Research has shown that there is a direct correlation between the diameter and the total dry bark yield, thus more data are not needed. The diameter of Deynap and Deykap were measured at 10 cm from the ground level, referred to as D10. The diameter of all stems more than 1 cm at D10 were recorded. Plants with a smaller diameter were recorded as < 1 cm. The diameters of multi-stemmed plants were converted into a single stem diameter. Based on this single stem diameter, the total dry bark weight were estimated based on a formula. Dead plants are not measured. A plant is declared dead, when it has no green leaves during vegetation time. If it is estimated that the plant has lost more than half of its bark, it is also not measured; if less than half of the bark is lost, the plant is measured as usual.

Data Analysis

Based on research carried out by C. Hadorn and P. Wangda (1999), a formula was developed to estimate the total dry bark yield of Deykap, *Daphne bholua* and *Daphne sureil* (there are no data available about the other *Daphne* species) for whole of Bhutan:

$$\mathbf{Wd = 0.02559 * D10^{2.27221}}$$

In which Wd is the estimated total dry bark weight in g, and
D10 is the diameter at 10 cm from the ground in mm

Based on this formula the total dry bark weight were estimated as a function of the diameter at 10 cm from the ground. According to the same research of C. Hadorn and P. Wangda (1999), some 47% of the total dry bark weight cannot be used and is considered as waste bark (for all diameters),

the remaining 53% is the usable bark. As per the calculations the total dry bark weight which can be collected is 7121644.98 g (7121.6 kg).

Table 4: Data analysis for *D.bholua*

Data analysis per block							
Block name:		Daphne Extraction Block					
Number of plants with a diameter of less than 1 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10<1cm	Average no of plants of D10<1cm per sample plot	Average no of plants of D10 < 1 cm per ha	Total no of plants of D10 < 1 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	539	7.7	770	58366	
Number of plants with a diameter of 1-1.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10: 1-1.9 cm	Average no of plants of D10: 1-1.9 cm per sample plot	Average no of plants of D10: 1-1.9 cm per ha	Total no of plants of D10: 1-1.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	206	3	294	22307	
Number of plants with a diameter of 2-2.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10:2-2.9 cm	Average no of plants of D10: 2-2.9 cm per sample plot	Average no of plants of D10: 2-2.9 cm per ha	Total no of plants of D10: 2-2.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	319	5	456	34543	
Available usable dry bark weight per sub-block of plants with diameter of > 3.0 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total Wd of plants with diameter of > 3 cm	Average Wd per sample plot	Average Wd per ha	Total Wd in sub-block	Total usable bark
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	(53% off)
				(d)	(e)	(f)	
D-Lagyel	75.8	70	124089	1772.7	177270	13437066	7121644.98
Note: Wd is the weight of bark in gram(g)							

Note: Wd is the weight of bark in gram(g)

People decide to follow the management prescriptions

- Cut the stem at 10 cm above the ground to promote regeneration of shoots through a coppice system. As a rule of thumb this is at the height of a fist with the thumb up. The bark is peeled off after the stem is cut.
- Cut the stem slanting to avoid rotting of the remaining stump
- Cut only stems of more than 3 cm in diameter at 10 cm from the ground; it is recommended to harvest as late as possible because the bark increment increases significantly over the years
- Take care and do not to destroy/ damage remaining stands, seedlings and saplings of Deynap and Deykap while harvesting

PART B: By-laws: Group Regulations and Norms

	Village	Number of member households	Gewog
	Goney-Wambur	63	Tsenkhar
	Total	63	
1. Members of the NWFP management & marketing group	Responsibilities and rights of members: • All the members shall be responsible for sustainable management of NWFPs. • The NWFPs shall be harvested during suitable season as prescribed in management plan. • The NWFPs group members shall follow and respect the instruction of executive members. • All members shall be vigilance for illegal harvesting of NWFPs by non-members and report to forest office/executive members. • All the group members have right to harvest NWFPs and market as per this plan. • The members will collect NWFPs as per the directives of the Chairperson based on demand of the trader/buyers but not exceeding the annual harvesting limit. • Members should inform the Chairperson/Forestry Official of any person extracting rural timber from NWFP harvesting area so as to stop the illegal harvesting of the NWFPs.		

2. Executive committee	
2.1 Chairperson	Name of Chairperson: Cheten Dorji
	Responsibilities of Coordinator/ Chairperson: • The chairperson has power to impose the fines and penalties if any members do not follow his instruction (meeting, timing). • Chairperson shall obtain the NWFPs collection permit from forest office during the collection season. • Inform or announce the date of NWFPs collection to each member. • Should coordinate the NWFPs harvesting. • Shall contact with the buyer/traders on the required quality of the NWFPs and inform the members accordingly. • Shall look for the buyers/traders. • To chair/coordinate all meetings related to this plan. • Resolve disputes and conflicts among members. • Take care of overall financial arrangements. • Attend/participate trainings/workshop/study tour when called.

2.2 Secretary	Name of Secretary: Jambay Dorji
	Responsibilities of Secretary: • Maintain/Keep the sale records. • Maintain the minutes of meeting. • Attain/participate trainings/workshop/study tour when called. • Assist the Chairperson. • Keep detail record of NWFPs harvest and submit to Chairperson

2.3 Treasurer	Name of Treasurer: Kinzang Tshomo
	Responsibilities of Treasurer: • Maintain cash transaction records. • Collect membership fees, incentives/compensation fees and deposit in bank • Look overall financial transaction in collaboration with the Chairperson and Secretary. • Officiate Secretary. • Keep detail record of NWFPs sale data. • Present account statement to the group during meeting • Issue money receipt after collection of fines and penalties

2.4 Resungpa	Name of Resuups: Geley Wangchuk, Nidup Wangdi, Sonamla, Tenzin Choiphel, Thinley Wangchuk
	Responsibilities of Resuups: • Patrol the NWFP area and report to Chairman.

3. Membership fee and other conditions	• Each member shall contribute Nu.150/- once in a year as a membership fee and shall be collected during the annual meeting. • Any new member shall pay 5% of the total amount collected till date • The fund will be deposited in the bank in the name of group counter sign by coordinator, secretary and treasurer. • Anyone wishing to withdraw from the group will have to put application stating all the reasons and no objection in future. The decision will be made during meeting to leave the group. • In unfavorable situation if one member wish to resign he/she be allowed with equal benefit sharing. • Once any member who had resigned from the group again wishing to join group will not be entertained. Any collection NWFP within the designated area is prohibited for them.
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4. Conflict resolution	• Conflicts or disputes within the NWFP Management Group, which cannot be resolved by the group members shall be forwarded to the Chairperson for settlement. • If the case cannot be resolved, it shall be forwarded to the Gewog Administration via Chiwog Tshogpa. • In case there is a conflict between the NWFP Group and outsiders, the case shall be forwarded to the Divisional Forest Office and Gewog Administration by the Chairperson.
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5. Penalties	• If any member sells the NWFPs directly to the traders without the notice of the group executive committee members shall be penalized as per the FNCRR, 2017. • If the non-members are found doing illegal/taking of NWFPs shall be dealt as per the FNCRR, 2017. • The group members shall sell the NWFPs to the valid licence holder only, if not the group shall be penalised as per the FNCRR, 2017. • The annual meeting shall be conducted on 30th month of December. Any members remaining absent in meeting should pay Nu.500/-as per the local village fine rules. • If the committee members fails to attend the meeting, he/she shall be levied a fine of Nu.700. The collected amount shall be deposited in the group saving account. However, during the times of death and sickness consideration will be made. • If any executive member's misuses the group fund, he/she has to pay double the amount within the date line given by the group.
6. Benefit sharing	• The group fund will be shared equally among the members • The group member are not allowed to collect NWFPs for individual marketing • The collection of NWFPs by outsider is strictly prohibited • The only small quantity is allowed to collect for self consumption

7. Authority of the Department	As per Chapter III, section 155 of FNCRR, 2017, the Department shall: • Suo moto or on the request by the Non Wood Forest Management Group (NWFMG) take action against violator of the Act, Rules or Management Plan; • Inspect, monitor, review the records of and otherwise supervise the activity of the NWFMG; and • Take action including cancellation of the NWFMG certificate if the NWFMG or any of its members who violate the provision of the Management Plan or any provision under this Chapter
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8. Cancellation of NWFP Management Certificate	As per Chapter III, section 156, 157 & 158 of FNCRR, 2017: • The certificate shall be cancelled, if the performance of NWFMG is not as per prescriptions of the approved management plan or the sustainability of the resource is not ensured • In the event the members of the NWFMG decide to cancel NWFP Management certificate in accordance with this Rules, the NWFP Management Certificate shall be deemed cancelled on the date of such decision being notified to CFO in writing and the NWFMG shall surrender the Certificate to the Department • The CFO shall have the right to enquire about the reason for such cancellation to ensure that no offence has been committed and shall notify the NWFMG on the cancellation of the certificate.
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Annexures I: NWFP Member List

No	Name	Village	Signature	Signature
1	Sonam Rinzhari	Lagay		
2	Tshewang Peldan	"		
3	Cheten Dorji	"		
4	Tendrel Zangmo	"		
5	Rinzin Tshomo	"		
6	Galay Tshering	"		
7	Kezang Tshering	Gongid		
8	Kuenzang Rangdrol	"		
9	Nidup Wangdi	"		
10	Sangay Wangdi	Trenchola		
11	Namgay Wangdi	Lagay		
12	Sonam Beki	Kiphu		
13	Toumo	"		
14	Yangdon	Dagorthang		
15	Beki Mo	"		
16	Tandin Wangmo	"		
17	Cheki Wangmo	"		
18	Pema Yangzom	Khimgong		
19	Sherab Pelmo	"		
20	Beki Yangzom	"		
21	Channing Harbu	"		
22	Beki Pelmo	"		

	Yuden Mo	Khingong	<u>ll</u>	<u>ll</u>
	Phuntsha Wangmo	"	<u>sgu</u>	<u>sgu</u>
4	Cheki Wangmo	Khablang	<u>ll</u>	<u>ll</u>
25	Jambay Dorji	"	<u>ll</u>	<u>ll</u>
26	Pema Lhamo	"	<u>ll</u>	<u>ll</u>
27	Tshomo	"	<u>ll</u>	<u>ll</u>
28	Norbu	"	<u>ll</u>	<u>ll</u>
29	Karma Cheki	"	<u>ll</u>	<u>ll</u>
30	Rinzin Mo	"	<u>ll</u>	<u>ll</u>
31	Yeshe Mo	"	<u>ll</u>	<u>ll</u>
32	Tanala	"	<u>ll</u>	<u>ll</u>
33	Kinlay Wangmo	"	<u>ll</u>	<u>ll</u>
34	Pema Choden	"	<u>ll</u>	<u>ll</u>
35	Tshering Dorji	"	<u>ll</u>	<u>ll</u>
36	Ugyen Wangmo	"	<u>ll</u>	<u>ll</u>
37	Ugyen Wangmo	Yanglong	<u>ll</u>	<u>ll</u>
38	Tsketen Lhamo	"	<u>ll</u>	<u>ll</u>
39	Rinchen Dama	"	<u>ll</u>	<u>ll</u>
40	Dorji Wangmo	"	<u>ll</u>	<u>ll</u>
41	Sonam Dama	Bara	<u>ll</u>	<u>ll</u>
42	Bumpa Mo	Gurtong	<u>ll</u>	<u>ll</u>
43	Shacha	Khablang	<u>ll</u>	<u>ll</u>
44	Tshering Yangki	"	<u>ll</u>	<u>ll</u>

	Jigme Thinley	Gajenshang	FD	FD
	Penca Tihoki	Godong		
47	Hangma	Jampang		
48	Pema Gyeltshen	"	my	my
49	Karma Tenzin	Sokorum		
50	Tashi Chopel	Sokchen		
51	Tihawang Pengju	"		
52	Lhacham Choden	Gang	LB	LB
53	Choden	"	Thul	Thul
54	Kinzang Tihomo	Jarkang		
55	Karma Chopel	"		
56	Sangay Gyeltshen	Chungka		
57	Thinlay Wangchuk	"		
58	Chimi Dorji	"	Thul	Thul
59	Sonam Chojay	"		
60	Jigme Singye	"		
61	Ugyen Dema	Beona	Thul	Thul
62	Tashi Wangma	Donajigbala		
63	Dorji Dema	Godong		

Annexure II: Block description for *Borinda grossa*

Block description form for <i>Borinda grossa</i>					
Date	26-28/01/2022				
Village	Lagyl	Name of the bamboo area	Rawapang Juug, Sadug Brangsa & Rashu Goenpa		
Block Name	Lagay Bamboo Production Area	Total area (ha)	285.8		
Access	What is the walking distance from the nearest village to the block				
	less than 3 hours	Within a day	More than a day		
	√				
	What is the walking distance from the block to the nearest road				
	less than 3 hours	Within a day	More than a day		
Slope	√				
	Very steep >100%	Steep 70-100%	Moderate 45-70	Gentle <45%	
Forest description	Dominant and associate species, canopy cover: Quercuss sp, Rhododendron sp				
Sub-Block	No	Density	Area (ha)	Productive area (%)	No. of sample plots
	1	Medium	94.5	34.4	5
	2	High	108	34.6	6
	3	Low	83.3	27.1	6
Bamboo flowering	Sporadic	Gregarious	Last time of the flowering	None	
	√		2007-2008		
Planation potential	Sub-blocks	Has plantation potential	No potential	Not applicable	
	1		√		
	2		√		
	3		√		
Current management	Current management practices in the entire block:				
Current threats	Threat	Description			
	Fire	No threat			
	Grazing	Mild			
	Illegal exploitation	Very less chance			
	Over exploitation	Less chance			
Opportunity	Future opportunity for the entire block: Can be used as mass bamboo production area and commercialization beyond locality				

Annexure III: Block description for *Daphne bholua*

Appendix 2: Block description form for Deynap and Deykap

Date: 4-14/2/2022

Village: Lagayel

Name of Deynap/ Deykap area: Deynap.

Block name: D-Lagayel

Total area (ha): 75

Access	What is the walking distance from the nearest village to the block?		
	Less than 3 hours ☒	Within a day	More than a day
	What is the walking distance from the block to the nearest road?		
	Less than 3 hours ☒	Within a day	More than a day

Slope	Very steep	Steep	Moderate	Gentle
	>100%	70-100%	45-70%	< 45 % ☒

Forest description	Dominant and associate species, canopy cover: Symplocus sp, Schima sp.
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Sub-block	No.	Density	Area (ha)	No of sample plots
	1		75	70
	2			
	3			

Plantation potential	Sub-blocks	Has plantation potential	No potential	Not applicable
	1	☒		
	2			
	3			

Current management	Current management practices in entire block: No management.
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Current threats	Threat	Description
	Fire	
	Grazing	☒
	Illegal exploitation	
	Over-exploitation	☒
Other:		

Opportunities	Future opportunities for the entire block: Plantation. and
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Annexure IV: *B. grossa* data analysis

Resource assessment compilation form for Borinda grossa						
Block name:	Lagay Bamboo Production Area	Dates of bamboo resource assessment:		26-28/01/2022		
Sub-block (Number)	Name of the sub-block	Area (ha)	% of productive area	Productive area (ha)	Number of sample plots	Remarks
1	Rewapang Juug	94.5	34.4	32.5	5	
2	Sadug Brangsa	108	34.6	37.4	6	
3	Rashu Goenpa	83.3	27.1	22.6	6	
Age class	Number of culms	Sub-block 1	Sub-block 2	Sub-block 3	Total in Block	
1 + year old bamboo culms	Total counted in sample plots	362	729	298	1389	
	Total harvestible in sample plots	218	460	208	886	
	Total in sub-block*	94120	181764	44899	320783	
	Total harvestible in sub-block*	56680	114693	31339	202712	
2 + years old bamboo culms	Total counted in sample plots	405	743	285	1433	
	Total harvestible in sample plots	294	492	179	965	
	Total in sub-block*	105300	185255	42940	333495	
	Total harvestible in sub-block*	76440	122672	26969	226081	
3 + years old bamboo culms	Total counted in sample plots	591	791	335	1717	
	Total harvestible in sample plots	111	266	119	496	
	Total in sub-block*	153660	197223	50473	401356	
	Total harvestible in sub-block*	28860	66323	17929	113112	
Total Stock in Sub-Block		353080	564241	138312	1055633	
Total Harvestible in Sub-Block		161980	303688	76237	541905	
AHL-33% (1 & 2 + years)		136360	245237	57757	439353	
AHL-70% (3+ years)		107562	138056	35331	280949	

Annexure V: *D.bholua* data analysis

Data analysis per block							
Block name:		Daphne Extraction Block					
Number of plants with a diameter of less than 1 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10<1cm	Average no of plants of D10<1cm per sample plot	Average no of plants of D10 < 1 cm per ha	Total no of plants of D10 < 1 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	539	7.7	770	58366	
Number of plants with a diameter of 1-1.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10: 1-1.9 cm	Average no of plants of D10: 1-1.9 cm per sample plot	Average no of plants of D10: 1-1.9 cm per ha	Total no of plants of D10: 1-1.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	206	3	294	22307	
Number of plants with a diameter of 2-2.9 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total no of plants of D10:2-2.9 cm	Average no of plants of D10: 2-2.9 cm per sample plot	Average no of plants of D10: 2-2.9 cm per ha	Total no of plants of D10: 2-2.9 cm in sub-block	
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	
				(d)	(e)	(f)	
D-Lagyel	75.8	70	319	5	456	34543	
Available usable dry bark weight per sub-block of plants with diameter of > 3.0 cm:							
Sub-block (name)	Area (ha)	No of sample plots	Total Wd of plants with diameter of > 3 cm	Average Wd per sample plot	Average Wd per ha	Total Wd in sub-block	Total usable bark
	(a)	(b)	(c)	(=c/b)	(=100*d)	(=a*e)	(53% off)
				(d)	(e)	(f)	
D-Lagyel	75.8	70	124089	1772.7	177270	13437066	7121644.98
Note: Wd is the weight of bark in gram(g)							

Annexure VI: Data Collection Pictures

